Department of Information and Communication Technology
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4th Year 1st Semester B. Sc. (Engg.) Final Examination 2025

Course Title: Data Mining | Course Code: ICT 4101
Time: 3 Hours | Marks: 70

Answer any 05 (FIVE) questions | Maintain sequence while answering

1. a) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
(i) What is the mode of the data? What is the midrange of the data? 2
(ii) Give the five-number summary of the data. 4

b) How is a quantile–quantile plot different from a quantile plot? 4

c) 'The median is one of the most important holistic measures in data analysis' – Is it true? State your argument. [ OR ] 4
Suppose that the values for a given set of data are grouped into intervals. The intervals and corresponding frequencies are as follows:

| age | frequency |
|---|---|
| 1–5 | 200 |
| 6–15 | 450 |
| 16–20 | 300 |
| 21–50 | 1500 |
| 51–80 | 700 |
| 81–110 | 44 |

Compute an approximate median value for the data.

2. a) What are the value ranges of normalization methods: (i) min-max normalization, (ii) z-score normalization, (iii) decimal scaling normalization. 4

b) Calculate (i) min-max and (ii) z-score normalization values of 35, for the numbers 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. 5

c) In real-world data, tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem. 5

3. A database has five transactions. Let, min_sup = 60% and min_conf = 80%.

| TID | items bought |
|---|---|
| T100 | {M,O,N,K,E,Y} |
| T200 | {D,O,N,K,E,Y} |
| T300 | {M,A,K,E} |
| T400 | {M,U,C,K,Y} |
| T500 | {C,O,O,K,I,E} |

a) Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes. 7

b) List all the strong association rules ( with support s and confidence c ) matching the following metarule, where X is a variable representing customers, and item i denotes variables representing items ( e.g., "A," "B," ): 7
$\forall x \in$ transaction, buys( X, item1 ) $\wedge$ buys ( X, item2 ) $\Rightarrow$ buys ( X, item3 ) [s,c]

| age | income | student | credit_rating | buys_computer |
|---|---|---|---|---|
| <=30 | high | no | fair | no |
| <=30 | high | no | excellent | no |
| 31…40 | high | no | fair | yes |
| >40 | medium | no | fair | yes |
| >40 | low | yes | fair | yes |
| >40 | low | yes | excellent | no |
| 31…40 | low | yes | excellent | yes |
| <=30 | medium | no | fair | no |
| <=30 | low | yes | fair | yes |
| >40 | medium | yes | fair | yes |
| <=30 | medium | yes | excellent | yes |
| 31…40 | medium | no | excellent | yes |
| 31…40 | high | yes | fair | yes |
| >40 | medium | no | excellent | no |

Table 1

| Actual Class\Predicted class | cancer = yes | cancer = no | Total |
|---|---|---|---|
| cancer = yes | 90 | 210 | 300 |
| cancer = no | 140 | 9560 | 9700 |
| Total | 230 | 9770 | 10000 |

Table 2



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4. (a) Define a Data Warehouse. State two key differences between an operational database (OLTP) and a data warehouse (OLAP). 2

(b) Draw a basic three-tier data warehouse architecture and briefly label its main components. 3

(c) Explain the multidimensional Data Model with a neat diagram. Describe the concepts of Data Cubes, Dimensions, and Fact Tables using a retail sales example (e.g. tracking sales by *Time*, *Item*, *Location*, and *Branch*). 6

(d) Consider a base data cube for a relation R( A, B, C ) with the base cardinality (number of distinct tuples) as follows: |ABC| = 40, |AB| = 20, |AC| = |BC| = 10, |A| = 10, |B| = 5, |C| = 8, and the apex/root |*All*| = 1. 3
(i) Calculate the total number of cuboids in the complete lattice for this 3-dimensional cube.
(ii) Compute the total number of aggregate units (size of the data cube) across all cuboids using the given carnalities.

5. a) Using the Naïve Bayes, find out the probability that a person less than 30 years of age, with medium income but a student with fair credit rating would not buy a computer. Use data from Table 1. 0.0067 6

b) Explain the terms (i) Accuracy, (ii) Sensitivity and (iii) Specificity. 4
For the Table 2, find (i) Accuracy, (ii) Sensitivity and (iii) Specificity values. 4
0.965 0.7 0.9956

6. a) Define (i) Entropy and (ii) Conditional Entropy. 4

b) From the Table 1, find out a decision for data ( 31...40, medium, yes, fair, ?). 6

c) Find *Info* ( Dj ) and Gain( age ) for the Table 1. 4

7. (a) Define the concept of outlier analysis in data mining. 2

(b) Distinguish between symmetric and asymmetric binary variables in the context of calculating data similarity. 2

(c) Explain the architectural workflow of the BIRCH clustering algorithm. Detail how the Clustering Feature (CF) tree helps handling large datasets efficiently. 5

(d) Given the following 1-dimensional dataset: X = { 2, 3, 4, 10, 11, 12, 20, 25, 30 } 5
(i) Manually apply the DBSCAN algorithm using the parameters Eps( radius ) = 2 and MinPts = 2.
(ii) Identify the core points, border points, and noise points.

8. (a) Define a data stream and state two main constraints of mining data structures. 2

(b) Differentiate between spatial data mining and multimedia data mining with one example each. 2

(c) Explain the process of Sequence pattern Mining in transactional databases. Discuss its significance, and describe how the Apriori-like property (downward closure property) is applied to find sequential patterns efficiently. 5

(d) Consider a tiny transactional database tracking customer item sequences with a minimum support threshold ($min_{sup}$) of 50% (i.e., absolute support count > 2). The database contains 4 sequences: 5

- Sequence 1: < ( a, b ), ( c ) >
- Sequence 2: < ( a ), ( b, c ) >
- Sequence 3: < ( a, b ), ( d ) >
- Sequence 4: < ( b ), ( c ) >

Find all frequent 1-sequence patters and calculate their support counts. Determine which of these 1-sequence qualify as frequent sequential patterns based on the $min_{sup}$ threshold.



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